Business Requirements Document

Hospital ER Performance Analysis

1. Purpose

This document outlines the business requirements for a Power BI project analyzing hospital emergency room (ER) performance. The goal is to give hospital stakeholders a clear, data-driven view of patient volume, wait times, satisfaction, and referrals, so they can make better decisions on staffing, resource allocation, and patient care.

2. Business Problem

Hospital management needs better visibility into how the emergency room is performing — how many patients are being seen, how long they wait, how satisfied they are, and how patients are being referred across departments. This project answers the question:

> *"How can the hospital use ER patient data to track performance, identify bottlenecks, and improve patient care and resource planning?"*

3. Objectives

- Track daily/monthly patient volume, wait times, satisfaction, and referrals.

Dashboard	Purpose
Monthly View	Monitor KPIs and trends for a selected month — admission status, age/gender/race distribution, department referrals, % seen within 30 mins, and patient volume by day & hour.

Consolidated View	Same metrics as Monthly View, aggregated over a custom date range for broader trend analysis.

- Identify peak days and hours to support staffing decisions.
- Understand patient demographics (age, gender, race) driving ER visits.
- Compare admitted vs. non-admitted patients and department referral patterns.
- Provide a patient-level detail view for granular investigation.
- Summarize key findings and recommendations for stakeholders.

4. Key KPI Requirements

KPI	Requirement
Number of Patients	Total daily patients visiting the ER, with a trend line to show peak days/seasonal patterns.
Average Wait Time	Average time before a patient is attended to, with a daily trend to flag highwait days.
Patient Satisfaction Score	Average daily satisfaction score, trended to spot dips linked to operational issues.
Patients Referred	Daily count of patients referred to other departments, trended to flag highreferral departments.

5. Dashboards to Deliver

Patient Details	Row-level grid of individual patients (ID, name, gender, age, admission date, race, wait time, department referral, admission status) for drill-down.
Key Takeaways	Written descriptive summary of patterns, anomalies, and recommendations across all dashboards.

6. Key Questions This Project Should Answer

- How many patients does the ER see, and how does that change month to month?
- What is the average wait time, and which days/hours are busiest?
- How satisfied are patients, and where does satisfaction drop?
- What share of patients are admitted vs. treated and released?
- Which age groups, genders, and racial groups make up most ER visits?
- Which departments receive the most referrals from the ER?

7. Tools & Data

Tools: Power BI (data modeling, DAX calculations, dashboard development).

Data: An ER patient dataset including patient ID, name, gender, age, race, admission date, admission status, wait time, department referral, and satisfaction score.

8. Assumptions & Constraints

- The dataset provided is a historical extract (April 2023 – October 2024), not a live feed.
- Wait time and satisfaction score are recorded per patient visit.
- Patient-identifying fields are used for internal analysis only and handled per hospital data privacy policy.

9. Success Criteria

- All four dashboards (Monthly View, Consolidated View, Patient Details, Key Takeaways) are built and functional.
- KPIs match the requirements in Section 4, with daily/monthly trend visuals.
- Dashboard is validated against known totals (e.g., total patients, admission split).
- Key Takeaways page delivers clear, actionable recommendations for ER operations.